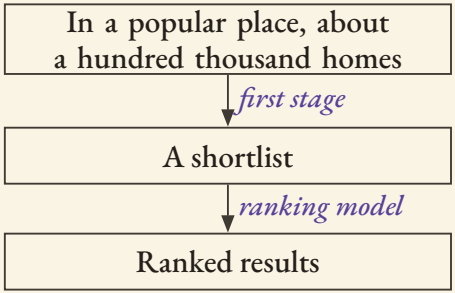
Airbnb: the first stage of search

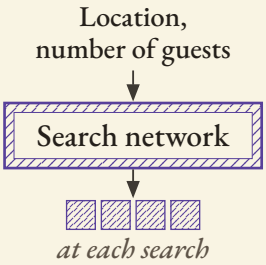
The first stage evaluates tens of thousands of homes.

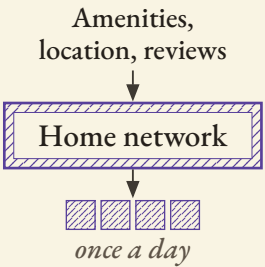
It used to be only a simple linear model that scored every home once a day, whatever the search.



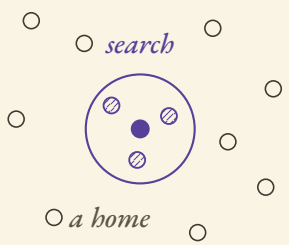
Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

Two networks and the nearest homes



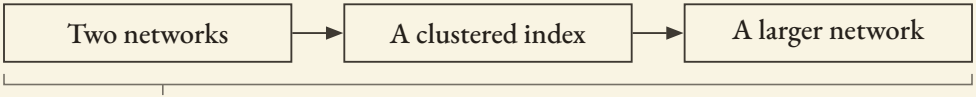


The nearest homes join the shortlist



Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

The value: three A/B tests



+0.31%

conversion, all three, relative change

Airbnb: “a large portion of the team’s annual goal”.

+2.3% *bookings from promotional emails*

−16% *compute with the clustered index*

Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.